

LAYER3D

Program: Layer3D version: 1.56
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Layer3d is a program to build .layer files for 3D simulations. In addition to all the capabilities of LayerBuilder to setup 2D .layer files of the devices, it also set up 3D device and simulation related properties. Layer3d read in .sol file, which contains information about the 3D structure and simulation of the device. The information of each 2D plane is recorded in the corresponding .layer files.

1. Overview of Layer3d

There are two view windows or zones in Layer3d for device layer and segments. The 2D-view zone displays 2D layers in the current z-segment, and the 3D-view zone displays the whole device.

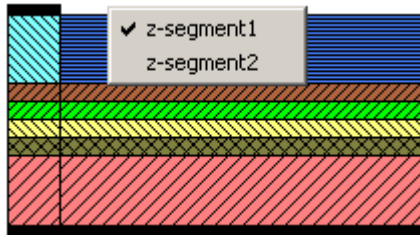


Figure 1: 2D-View Zone

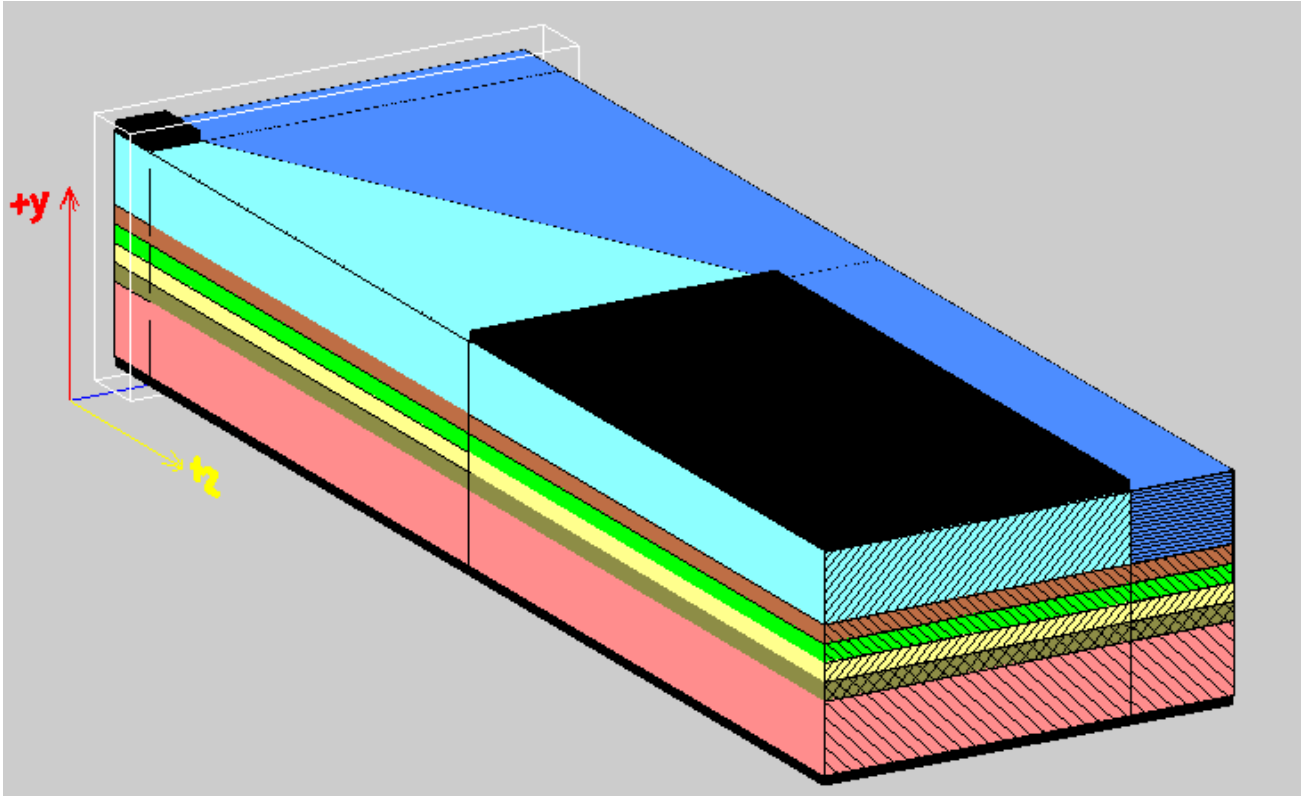
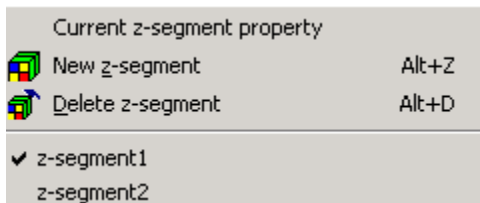


Figure 2: 3D-View Zone

2. Layer3d menu

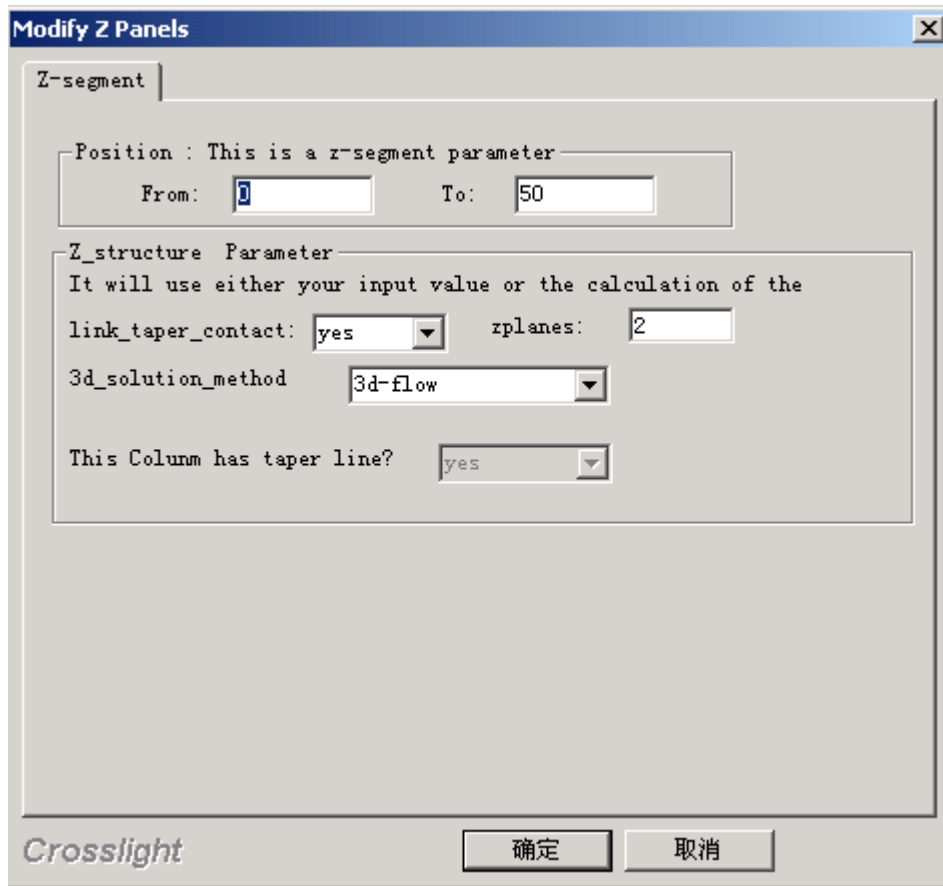
Following are menus to setup the segment in z-direction. The menus used to set 2D .layer file (xy plane) is similar to the LayerBuilder. Please refer to LayerBuilder help document for their usage.

(2.1) Z-segment menu



(2.1.1) Current z-segment property

Set properties of current segment. Following is the dialog box that pops up when “Current z-segment property” menu is clicked.



Modify Z Panels

Z-segment

Position : This is a z-segment parameter

From: To:

Z_structure Parameter

It will use either your input value or the calculation of the

link_taper_contact: zplanes:

3d_solution_method

This Column has taper line?

Crosslight

From, To: start and end coordinates in z-direction.

Line_taper_contact: set related values for the contact between segments. For details of this property, please refer to the crosslight reference manual.

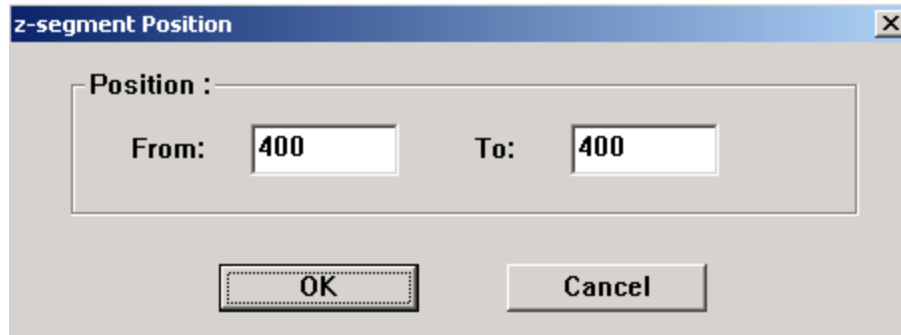
zplanes: number of mesh in z-direction of the current segment. Note: This value is effective when “3d_solution_method” is set to “3d_flow”.

3d_solution_method: the 3D simulation method used in the core simulator.

This Column has taper line?: set if taper line exist or not. Refer to corresponding document of the core simulator. To specify the column, click a layer in the corresponding column with mouse.

(2.1.2) New z-segment

Create and set up a new segment.



From, To: Start and end coordinates of the segment in z-direction.

(2.1.3) Delete z-segment: delete current z-segment.

(2.1.4) z-segment1, z-segment2... : segments existing in device. User can choose a segment and edit it. The marked segment is the current active segment.

(3) New icons in the tool bar in Layer3d



These are icons added to the tool bar of Layer3d in this version. They are used for the 3D view of the device. Choose 3D view window or zone to enable these icons.

(3.1) : parallel view of the 3D object.

(3.2) : perspective view of the 3D object.

(3.3) : view point distance in the perspective view.

5 Three-dimensional Examples

5.1 Example 1

Location: *crosslig\apsys_examples\3d_flow\taper_3f_a*

1. Create a new z-segment

- Create the first column

l) Create the first layer of the first column. There are three ways to construct a new z-segment:

- <alt+z> (hold both "alt" and "z" keys).
- Menu "z-segment" - new z-segment, or
- the "new z-segment" icon on the toolbar.

A dialog box named “New Segment” will pop up. Set the “To:” field to 20, leave the other parameters to the default, and press “OK”. This will automatically create a column and layer, as well.

Right click on the new layer in the 2D view to bring up the “Modify Layer” dialog box. Input the info for the current layer (i.e. the bottom layer of the first column) which is listed below:
set size_info: height 1.35, width 1

Modify Layer

Layer_Mesh DFB/DBR grating setting Vcsl

Size_Info Material Doping Contact Column_Mesh

Height:

The Height Of Current Layer 1.35 uM

Width:

The Width Of Current Layer 1 uM

Convert To Irregular Graph

Upper_W#(The Width Of Current Layer's Upper Edge) 6 uM

Lower_W#(The Width Of Current Layer's Lower Edge) 6 uM

Using this layer to solve wave equation?

yes

Crosslight OK Cancel

Figure 3: Set Size Info

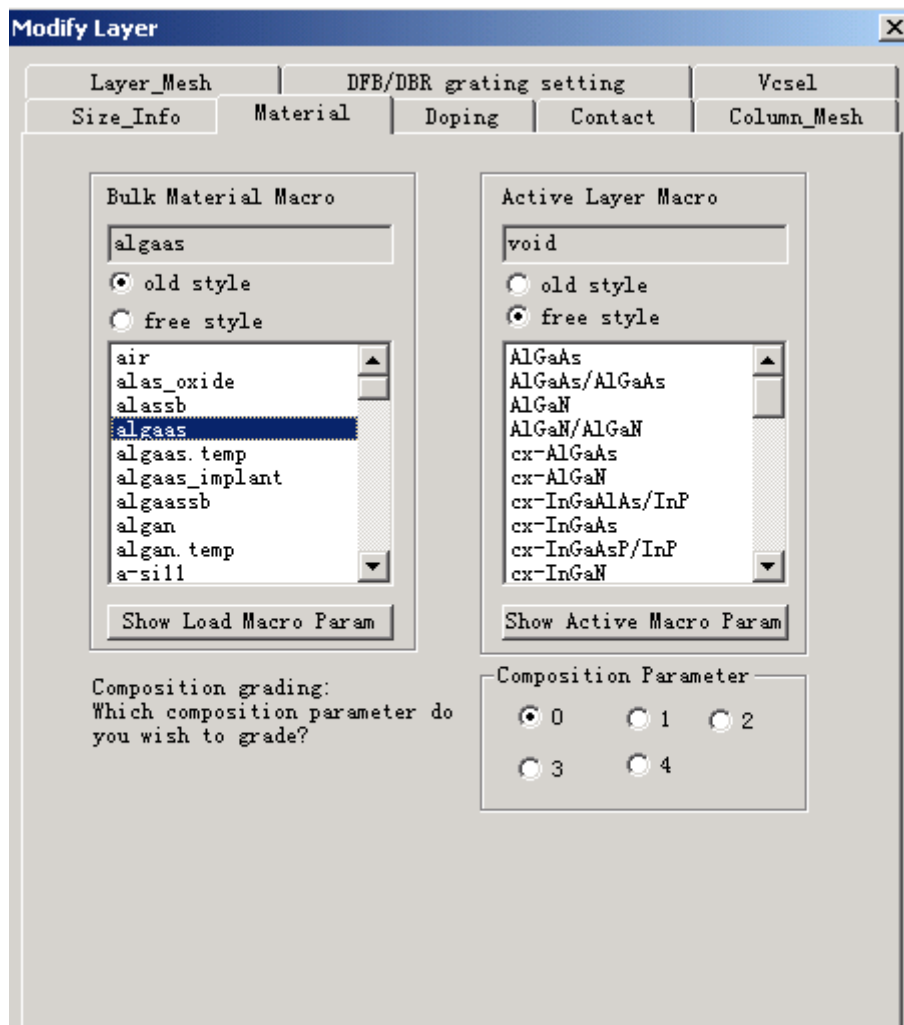


Figure 4: Set Material Property

Set material: double click "algaas" in the field of bulk material and input 0.71 in the dialog "algaas." (Note: make sure the radio button 0 is checked at "parameter composition" field.)

Modify Layer

Layer_Mesh	DFB/DBR grating setting		Vcsel
Size_Info	Material	Doping	Column_Mesh

Define doping profile

Doping type:

☐ NONE

☒ N

☐ P

Concentration (m^{-3}):

Gauss Tail ($\mu M \neq 0$):

n_newdoping_index:

p_newdoping_index:

Figure 5: Set Doping Property

Set doping: choose "n" (n type doping) and input "1e24" for its concentration.

Layer_Mesh	DFB/DBR grating setting		Vcsl		
Size_Info	Material	Doping	Contact	Column_Mesh	
	Side	From	To	Type	Link Prev Seg
Contact 1	top	0	1	ohmic	FALSE
Contact 2	top	0	1	ohmic	FALSE
Contact 3					
Contact 4					
Contact 5					
Contact 6					
Contact 7					
Contact 8					

Notes: Double click grid to select contact type!

Figure 6: Set Contact Property

set contact:

- side - the contact's location, here choose "bottom"
- from - the start point of the contact, set it to "0"
- to - the end point of the contact, set it to "1"
- type - contact type, choose "ohmic"
- Link Prev Seg - whether or not to connect this contact to the one in the previous z-segment. Here we set it to "false" (default value).

Modify Layer

Layer_Mesh DFB/DBR grating setting Vcsel

Size_Info Material Doping Contact Column_Mesh

Please input the mesh infomation of this column

Mesh Number: 4

Mesh Ratio: 0.8

Shift Center (micron): 0

Auto Mesh (Optional)

Total_X_Mesh: 0

Total_Y_Mesh: 0

Figure 7: Set Column Mesh

set column_mesh:

- mesh number 4
- mesh ratio 0.8
- shift centre 0,
- extra_mesh_left "no"
- extra_mesh_right "no"
- total_X_mesh 0, total_y_mesh 0
(default value, if you want to use the "auto mesh" function, you may need to edit two values)

The image shows a software window titled "Modify Layer" with a close button (X) in the top right corner. The window contains several tabs: "Size_Info", "Material", "Doping", "Contact", "Column_Mesh", "Layer_Mesh", "DFB/DBR grating setting", and "Vcsl". The "Layer_Mesh" tab is currently selected. Below the tabs, there is a text prompt: "Please input the mesh infomation of this layer". Under this prompt, there are three input fields, each with a label and a text box: "Mesh Number" with the value "5", "Mesh Ratio" with the value "0.8", and "Shift Center (micron)" with the value "0".

Size_Info	Material	Doping	Contact	Column_Mesh
Layer_Mesh	DFB/DBR grating setting	Vcsl		

Please input the mesh infomation of this layer

Mesh Number: 5

Mesh Ratio: 0.8

Shift Center (micron): 0

Figure 8: Set Layer Mesh

set layer_mesh:

- mesh number 5
- mesh ratio 0.8
- shift centre 0
- extra_mesh_upper "no"
- extra_mesh_lower "no"
- Click "ok" to finish this layer.

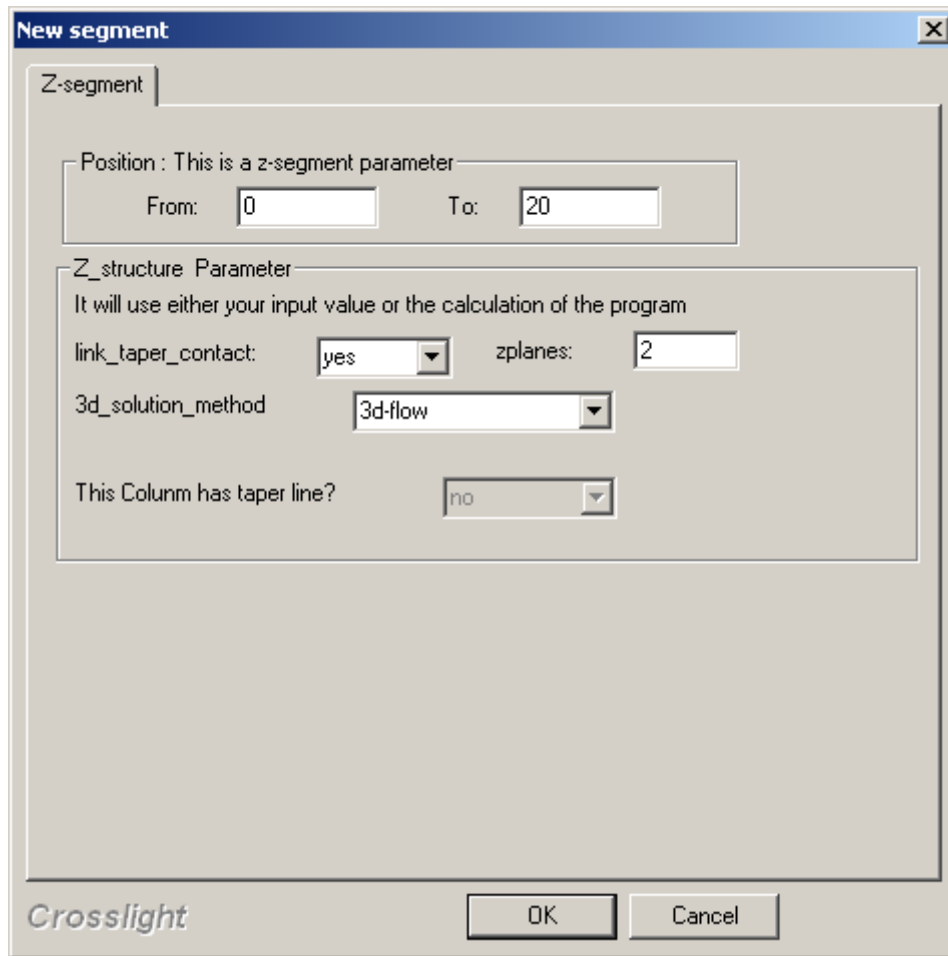


Figure 9: Set Z-Segment Property

set z-segment (use the Z-Segment->Current Z-Segment Property menu item):

- z-planes 2

II) Go on to the second layer.

Select the layer we have created and click the icon "new layer" to pop up the dialog box, "New Layer."

set size_info: height 0.1462, width 1

set material: double click "algaas" in the field of bulk material, click "ok" and then check radio button 1 at "parameter composition" field.

In the dialog box of composition_grading, choose "+y" for direction

Input 0.71 for grading_from, and 0.33 for grading_to. Then click "ok"

set doping: choose "none".

set contact: do nothing.

set column_mesh: do nothing.

set layer_mesh: mesh number 3,
mesh 0.8,
shift centre 0,

set other: let it be null

set z-segment: do nothing.

Now the second layer is finished.

III) The third layer

Select the second layer and click the icon "new layer" to pop up the dialog box, "New Layer"

set size_info: height 0.0076, width 1

set material: double click "gaas" in the field of bulk material.

- Double click "AlGaAs/AlGaAs" in the active macro field.
- Input 0 for xw
0.33 for xb
300 for temperature.
- Click "ok"

set layer_mesh: mesh number 3

- mesh ratio 1
- shift centre 0

Do nothing with the remaining settings.

IV) The fourth layer

Select the 3rd layer and click the icon "new layer" to pop up the dialog box, "New Layer"

set size_info: height 0.1462; width 1

set material: double click "algaas" in the field of bulk material and click "ok"

- Check radio button 1 at "parameter composition" field.
- In the dialog box of composition_grading, choose "+y" for direction.
- Input 0.33 for grading_from, and 0.71 for grading_to.
- Click "ok"

set layer_mesh: mesh number 3

- mesh 1.2
- shift centre 0

Do nothing with the remaining settings.

V) The 5th layer

Select the 4th layer and click the icon "new layer" to pop up the dialog box, "New Layer"

set size_info: height 0.0538; width 1

set material: double click "algaas" and input 0.71, then click "ok"

set layer mesh: mesh number 3,

- mesh 1.2
- shift centre 0,

Do nothing with the remaining settings.

VI) The 6th layer

Select the 5th layer and click the icon "new layer" to pop up the dialog box "New Layer"

set size_info: height 1.2962; width 1

set material: double click "algaas" and input 0.71, then click "ok"

set layer_mesh: mesh number 5

- mesh 1.2
- shift centre 0,

set doping: choose "p" (p type doping) and input "1e24" for its concentration.

set contact: side - "top"

- input 0 for start point and 1 for end point
- Use "ohmic" for contact type

Do nothing with the remaining settings.

Now the first column is completed.

2. Create the second column:

- The second column is almost the same as the first one. We can use the "copy column" function as follows.

I) Select one layer of the first column and click the icon "copy column" from the toolbar. Then click "paste column" from the toolbar; some modifications are needed as shown in the following steps.

II) Select the bottom layer of this column and edit it:

- set size_info: width 7
- set column_mesh: use 5 for mesh number and 1.2 for mesh ratio
- set contact: side - "bottom"
 - from - "0"
 - to - "7"
 - type - "ohmic"

III) Select the top layer of this column

- set material: double click "vacuum" for its bulk material
- set doping: select "none" for doping type

For now, the first z-segment has been completed.

3. Create the second segment

Click the "new z-segment" icon in the toolbar, a dialog box "z-segment position" will pop up requesting the position for this segment. Input 200 for "from" and 400 for "to." Then click "ok."
Some modifications are needed with this z-segment.

I) The columns' widths are different from those of the first z-segment.

- Select the bottom layer of the first column and change its width to 6.
- Select the bottom layer of the second column and change its width to 2.
- In set column_mesh, modify mesh number to 4.

II) This segment has a different tape_point than that of the first z-segment.

- Select one layer of this z-segment and choose: set z-segment.

III) The contacts of this z-segment are different from those of the first z-segment.

- For the first column, select contact setting of the bottom layer:

- side - "bottom"
- from - "0"
- to - "6"
- type - "ohmic"
- Link Prev Seg - "true"

Then select the contact setting of the top layer:

- side - "top"
- From - "0"
- To - "6"
- type - "ohmic"

For the second column, select contact setting of the bottom layer:

- side - "bottom"
- from - "0"
- to - "2"
- Link Prev Seg - "true"

IV) Column mesh allocation

The mesh number for the first column is 5.

The mesh number for the second column is 4.

To edit the mesh number of each column, select the bottom layer of that layer and check column_mesh setting to edit it.

Now the device setup is completed. You can choose menu "File" -> save and input the name "a" to save it. Three files "a.layer", "a1.layer" and "a.sol" will be generated.

3.2 Example 2

Location: *crosslig\apsys_examples\3d_flow\taper_3f_b*

The example is similar to Example 1, except that a segment with one z-plane is inserted between the existing z-segments. We can use the input files from the previous example to build the input files of this example so that we do not need to start from scratch.

1. Open the previous file "a.sol"
2. Choose z-segment2 and click icon "delete z-segment" to delete it.
3. Click icon "new z-segment" to add a new segment with one z-plane.
A dialog pops up to request the position.
4. Input 110 for "from" and 110 for "to" in the boxes.
5. Edit this segment according to the device. Make sure this segment, i.e. z-segment, is chosen (right click the mouse in the blank area to check.)

a) Edit z-segment parameter

- Right click any layer of the first column and select z-segment position.
- Input from "110", and to "110"
- Input z-planes "1"

Since there is no taper line in the second column, there is no need to deal with it for the 2nd column.

b) Edit widths of the two columns

- Change the width of the first column to 3.5 micron.
- Change the width of the second column to 4.5 micron.

c) Edit contact in this segment

Because the bottom of this device has a connected uniform contact, the contact in this segment should be connected to the previous z-segment. Select the contact setting of the bottom layer of the first column:

- side - "bottom"
- from - "0"
- to - "3.5"

- type - "ohmic"
- "Link Prev Seg" - "true"

Select the contact setting of the top layer of the first column:

- side - "top"
- from - "0"
- to - "3.5"
- type - "ohmic"

Select the contact setting of the bottom layer of the second column:

- side - "bottom"
- from - "0"
- to - "4.5"
- type - "ohmic";
- Link Prev Seg - "true".

d) column_mesh allotment

- The mesh number for the first column is 4.
- The mesh number for the second column is 4.

2. Go on with the third segment

Click the "new z-segment" icon to add a segment. A dialog box pops up to request the position. Input 200 for "from" and 400 for "to" in the boxes.

Edit this segment according to the device.

a) z-segment parameter

- Right click any layer of the first column
- Select z-segment position from "200" to "400"
- z-planes "2",
- Since there is no taper line in the second column, there is no need to deal with it here.

b) Widths of the two columns:

- Change the width of the first column to 6 microns.
- Change the width of the second column to 2 microns.

c) Contacts in this segment

Because the bottom of the device has a connected contact, the contact should connect to the previous z-segment.

Select the contact setting of the top layer of the first column:

- side - "top"
- from - "0"
- to - "6"
- type - "ohmic"
- Link Prev Seg - "false"

Select the contact setting of the bottom layer of the first column:

- side - "bottom"
- from - "0"
- to - "6"
- type - "ohmic"
- Link Prev Seg - "true"

Select the contact setting of the bottom layer of the second column:

- side - "bottom"
- from - "0"
- to - "2"
- type - "ohmic"
- Link Prev Seg - "true"

d) Column mesh allotment

- The mesh number for the first column is 5.
- The mesh number for the second column is 4.

Now the device has been completed. Choose menu "File" -> "save" and input "b" for its name to save it. Four files named "b.layer", "b1.layer", "b2.layer" and "b.sol" are generated.

3.3 Example 3

Location: *crosslig\apsys_examples\3d_flow\led3d_thermal*

1. Create the first z-segment of the device

a) Create the first layer of the first column:

set size info: height 2; width 400
set material : bulk material - gan
set doping : type - n; concentration - 5e24
set column mesh : mesh number - 8; mesh ratio - 0.85
set layer mesh : mesh number - 8; mesh ratio - 0.8
set z-segment: z-position, from "0" to "100"; zplanes - 2;

Do nothing with the remaining settings.

b) The second layer:

Copy and paste the first layer.
set layer mesh: mesh number - 8
• mesh ratio - 0.9

Do nothing with the remaining settings.

c) The third layer:

set size info : height 0.003; width 400
set material : bulk material - ingan, "in" composition - 0.4
active material- InGaN/InGaN, xw - 0.4, yw - 0.08, temperature - 300
set doping : type - none
set layer mesh : mesh number - 3
• mesh ratio - 1

Do nothing with the remaining settings.

d) The fourth layer:

set size info : height 0.005; width 400
set material : bulk material - ingan, "in" composition - 0.08
set layer mesh : mesh number - 3
• mesh ratio - 1

Do nothing with the remaining settings.

e) Copy the third layer to the 5th, 7th, 9th and 11th layers.

f) Copy the fourth layer to the 6th, 8th, 10th and 12th layers.

- g) The 13th layer
set size info : height 0.1; width 400
set material : bulk material - gan
set doping : type - p, concentration - 2e25
set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

- h) The 14th layer
set size info : height 0.1; width 400
set material : bulk material - gan
set doping : type - p, concentration - 2e25
set layer mesh : mesh number - 3
- mesh ratio - 1
- set contact : side - "top", from - 0, to - 390, type - "ohmic"
Do nothing with the remaining settings.

The first column is completed. We need to copy it to the second column.

- a) Keep the bottom layer and cut the other layers of this column.
- b) Set size info, column mesh and contact for this column:
set size info: width - 140
set column mesh: mesh number - 5, mesh ratio - 1.15
set contact: side - "top", from - "20", to - "140", type - "ohmic"

2. Create the second z-segment

- a) Specify the z-segment parameters
position: from "150", to "400"
zplanes: 2
- b) Set top contact for the first column: Link Prev Seg - "true"

Remove top contact of the second column by deleting any box of the contact. For example, you can delete the value "top" in box "side."

Now that the device setup is completed, save its name as "ganled."
Three files named "ganled.layer", "ganled.layer" and "ganled.sol" are generated.

4 Two-dimensional Examples

This section demonstrates the setup of a 2D layered device using Layer3D. Two-dimensional examples can also be built with another Crosslight GUI product, LayerBuilder.

4.1 Example 4

Examples for Lastip: crosslig\lastip_examples\A_layered_LD

It is a simple device. Use "drag and release" function in this example.
Drag and release skill: Drag the source layer to be copied onto the destination layer, release when the destination layer has turned grey.

1. Use the means mentioned in 3d examples to create the bottom layer.
Input the parameters including height, width, layer_mesh, doping, column_mesh and bottom contact.
set size info: height - 1.35, width - 1.5
set material: bulk material - algaas, "al" composition - 0.71
set doping: doping type - n, concentration - 1e24
set contact: side - "bottom", from - 0 , to - 1.5, type - "ohmic"
set layer mesh : mesh number - 9
 - mesh ratio - 0.8set column mesh: mesh number - 2, mesh ratio - 1
Do nothing with the remaining settings.
2. Create the second layer:
set size info : height - 0.1462, width - 1.5
set material : bulk material - algaas
 - composition parameter - 1
 - direction - "+y"
 - grading_from - 0.71
 - grading_to - 0.33set layer mesh: mesh number - 6
 - mesh ratio - 0.8Do nothing with the remaining settings.
3. Create the third layer - active layer
set size info : height - 0.0076, width - 1.5
set material : bulk material - gaas
 - active macro - AlGaAs/AlGaAs
 - xw - 0
 - xb - 0.33
 - temperature - 300set layer mesh : mesh number - 4
 - mesh ratio - 1Do nothing with the remaining settings.
4. Create the fourth layer:
copy the second layer and paste here.
set material : bulk material - algaas
 - composition parameter - 1
 - direction - "+y"
 - grading_from - 0.33
 - grading_to - 0.71set layer mesh : mesh number - 6
 - mesh ratio - 1.2Do nothing with remaining settings.
5. Use drag and drop to copy the bottom layer and paste it to this layer.
set doping: doping type - p , concentration - 1e24
set layer mesh: mesh number - 9, mesh ratio - 1.2
set contact : side - "top", from - 0, to - 1.5, type - "ohmic"

Do nothing with remaining settings.

6. Now we can save it to a .layer file (not .sol as in the 3-d case)

Choose menu "File" -> "save as"

A dialog "save as" pops up. In the dialog select "Layer Files (*.layer)" for its file type, and input its name

"crosslig\lastip_examples\A_layered_LD\gaas10"

Until now the device is completed and a file named gaas10.layer is saved.

4.2 Example 5

Examples for Lastip: crosslig\lastip_examples\current_blocking\leaky_mqw

Irregular polygons will be drawn in this example. This example consists of three columns, so "copy column" and "paste column" functions will be applied. Make sure to edit the properties of the newly copied column, such as "column mesh," width, contact, etc.

There is a switch button "convert to irregular/regular graph" located at the lower part of set size_info, which is used to draw irregular polygons.

1. Create the first column:

You can refer to the file "yoko2.mater" to find out the material macros for m1, m2, m3 and file "yoko2.doping" for doping profile of each layer.

a) The first layer

set size info : height - 0.5, width - 1

set material : bulk material - inp

set doping : doping type - n, concentration - 5e24

set contact : side - "bottom", from - 0 , to - 1, type - "ohmic"

set layer mesh : mesh number - 5, mesh ratio - 0.8

set column mesh : mesh number - 5

- mesh ratio - 0.8

Do nothing with the remaining settings.

b) The 2nd layer:

set size info : height - 0.36, width - 1

set material : bulk material - inp

set doping : doping type - n, concentration - 5e24

set layer mesh : mesh number - 5, mesh ratio - -1.2

Do nothing with the remaining settings.

c) The 3rd layer:

set size info : height - 0.12, width - 1

set material : bulk material - inp

set doping : doping type - n, concentration - 6e23

set layer mesh : mesh number - 5, mesh ratio - -1.2

Do nothing with the remaining settings.

d) The 4th layer:

copy from the 3rd layer

e) The 5th layer:

copy from the 3rd layer

set layer mesh : mesh number - 5

- mesh ratio - 1.2

f) The 6th layer

set size info : height - 1.2, width - 1

set material : bulk material - ingaalp; y - 0.44

set doping : doping type - n, concentration - 6e23

set layer mesh : mesh number - 6

- mesh ratio - 1

Do nothing with the remaining settings.

g) The 7th layer:

set size info : height - 0.29, width - 1

set material : bulk material - inp

set doping : doping type - n, concentration - 6e23

set layer mesh : mesh number - 5

- mesh ratio - 1

Do nothing with the remaining settings.

h) The 8th layer

set size info : height - 1.2, width - 1

set material : bulk material - ingaalp; y - 0.44

set doping : doping type - n, concentration - 1e23

set layer mesh : mesh number - 6

- mesh ratio - 1

Do nothing with the remaining settings.

i) The 9th layer:

set size info : height - 0.01, width - 1

set material : bulk material - ingaalp

- y - 1
- active macro - InGaAsP/InP
- xw - 0.467
- yw - 1
- xb - 0.202
- yb - 0.44
- temperature - 300

set doping : doping type - n, concentration - 1e23

set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

j) The tenth layer

copy the 8th layer and paste here.

set size info : height - 0.1 , width - 1

Do nothing with the remaining settings.

k) The 11th layer:

copy from the 9th layer

l) The 12th layer:

copy from the 10th layer

m) The 13th layer:

copy from the 9th layer

n) The 14th layer:

set size info : height - 1.2, width - 1

set material : bulk material - ingaasp; y - 0.44

set doping : doping type - n, concentration - $1e23$

set layer mesh : mesh number - 8

- mesh ratio - 1

Do nothing with the remaining settings.

o) The 15th layer:

set size info : height - 1.2, width - 1

set material : bulk material -inp

set doping : doping type - p, concentration - $5e23$

set layer mesh : mesh number - 6

- mesh ratio - 1

Do nothing with the remaining settings.

p) The 16th layer:

set size info : height - 0.5, width - 1

set material : bulk material -inp

set doping : doping type - p, concentration - $5e23$

set layer mesh : mesh number - 4, mesh ratio - 1

set contact : side - "top", from - 0, to - 1, type - "ohmic"

Do nothing with the remaining settings.

2. After the first column is finished, we can use the functions "copy column" and "paste column" in either the menu or toolbar. Modifications are needed with some layers. The layer that under the top layer is not a rectangle, so edit it in "no-irregular graph" mode.

a) The first layer:

set size info : height - 0.5, width - 0.2

set column mesh : mesh number - 5

- mesh ratio - -1.3

set contact : side - "bottom", from - 0, to - 0.2, type - "ohmic"

Do nothing with the remaining settings.

b) From the 2nd layer to the 13th layer

set material: bulk material - inp

set doping : doping type - p. concentration - $1e24$

Do nothing with the remaining settings.

c) The 14th layer:

set material: bulk material - inp

set doping : doping type - p. concentration - $5e23$

Do nothing with the remaining settings.

d) The 15th layer:

set size info: click button "convert to irregular graph" upper_w# - 0.8, lower_w# - 0.2

set material: bulk material - inp

set doping : doping type - p. concentration - $5e23$

Do nothing with the remaining settings.

e) The 16th layer:

set size info: click button "convert to regular graph" height - 0.5, width - 0.8

set material: bulk material - inp

set doping : doping type - p. concentration - $5e23$

set contact : side - "top", from - 0, to - 0.8, type - "ohmic"

Do nothing with the remaining settings.

3. Create the third column. Copy the first column and paste here

a) The first layer:

set size info : height - 0.5, width - 1.8

set column mesh : mesh number - 5, mesh ratio - 1.4

set contact : side - "bottom", from - 0, to - 1.8, type - "ohmic"

Do nothing with the remaining settings.

b) From the second layer to the 13th layer:

set material: bulk material - inp

set doping : doping type - p. concentration - $1e24$

Do nothing with the remaining settings.

c) The 14th layer:

set material: bulk material - inp

set doping : doping type - n. concentration - $2e24$

Do nothing with the remaining settings.

d) The 15th layer:

set size info: click button "convert to irregular graph" upper_w# - 1.2, lower_w# - 1.8

set material: bulk material - inp

set doping : doping type - n. concentration - $2e24$

Do nothing with the remaining settings.

e) The 16th layer:

set size info: click button "convert to regular graph" height - 0.5, width - 1.2

set material: bulk material - inp

set doping : doping type - p. concentration - $5e23$

set contact : side - "top", from - 0, to - 1.2, type - "ohmic"

Do nothing with the remaining settings.

f) Save it as "crosslig\lastip_examples\current_blocking\leaky_mqw\yoko0.layer"

4.3 Example 6

Examples for Lastip: crosslig\lastip_examples\broad_areas\broad_area2

For the statements "include file= " in this example, simply find the included files and refer to the parameters.

1. Create the first column.

You need to find the parameters and their values in files: "ox.wel" and "ox.bar."

a) Create the first layer:

set size info : height - 2, width - 20
set material : bulk material - gaas
set doping : doping type - n, concentration - $2e24$
set contact : side - "bottom", from - 0 , to - 20, type - "ohmic"
set layer mesh : mesh number - 6, mesh ratio - 0.8
set column mesh : mesh number - 7, mesh ratio - 0.7

Do nothing with the remaining settings.

b) The second layer:

set size info : height - 1.75, width - 20
set material : bulk material - ingaalp, x - 0.7
set doping : doping type - n, concentration - $1e24$
set layer mesh : mesh number - 6, mesh ratio - 0.8

Do nothing with the remaining settings.

c) The 3rd layer:

set size info : height - 0.1, width - 20
set material : bulk material - ingaalp, x - 0.5
set doping : doping type - n, concentration - $2e21$
set layer mesh : mesh number - 4
• mesh ratio - 0.9

Do nothing with the remaining settings.

d) The 4th layer

set size info : height - 0.007, width - 20
set material : bulk material - ingap
• x - 0.42
• active macro - InGaAlP/GaAs
• xw - 0.51
• yw - 0
• xb - 0.25
• yb - 0.25
• temperature - 300
set doping : doping type - n, concentration - $2e21$
set layer mesh : mesh number - 3
• mesh ratio - 1

Do nothing with the remaining settings.

e) The 5th layer:

copy the 3rd layer and paste here.
set size info : height - 0.01, width - 20

Do nothing with the remaining settings.

f) The 6th layer:

copy the 4th layer and paste here.

g) The 7th layer:

copy the 5th layer and paste here.

h) The 8th layer:

copy the 6th layer and paste here.

i) The 9th layer:

copy the 7th layer and paste here.
set size info : height - 0.1, width - 20
set layer mesh : mesh number - 4
• mesh ratio - 1.1

Do nothing with the remaining settings.

j) The tenth layer:

set size info : height - 1.75, width - 20
set material : bulk material - ingaalp, x - 0.7
set doping : doping type - p, concentration - 1e24
set layer mesh : mesh number - 6
• mesh ratio - 1.2

Do nothing with the remaining settings.

k) The 11th layer:

set size info : height - 0.05, width - 20
set material : bulk material - ingaalp, x - 0.51
set doping : doping type - p, concentration - 3e25
set layer mesh : mesh number - 4
• mesh ratio - 1.3

Do nothing with the remaining settings.

l) The 12th layer:

set size info : height - 0.2, width - 20
set material : bulk material - gaas
set doping : doping type - p, concentration - 1e26
set layer mesh : mesh number - 3
• mesh ratio - 1.3

Do nothing with the remaining settings.

m) The 13th layer:

copy the 12th layer and paste here.
set layer mesh : mesh number - 4, mesh ratio - 1.3, extra_mesh_lower - yes
set contact : side - "top", from - 0, to - 20, type - "ohmic"

Do nothing with the remaining settings.

2. Copy the first column to create the second one.

a) Change the width, column mesh, mesh ratio, etc., of this column by right-clicking the bottom layer.

set size info : height - 2, width - 80
set contact : side - "bottom", from - 0 , to - 80, type - "ohmic"
set column mesh : mesh number - 9, mesh ratio - 1.3

Do nothing with the remaining settings.

b) Change the bulk macro of the top layer and remove the top contact.

set material : bulk macro - sio2
remove "top" contact

c) Remove doping profile of every layer of this column.

3. Save it as file "crosslig\lastip_examples\broad_areas\broad_area2\ox.layer"

4.4 Example 7

Examples for Lastip: crosslig\lastip_examples\short_wavelength\ingaalp_650nm

There are several non-rectangle polygons in this example. When you try to copy columns, you should be very careful.

1. Create the first column.

a) Create the first layer:

```
set size info : height - 5, width - 2.5
set material : bulk material - gaas
set doping : doping type - n, concentration - 8e24
set contact : side - "bottom", from - 0 , to - 2.5, type - "ohmic"
set layer mesh : mesh number - 6, mesh ratio - 0.8
set column mesh : mesh number - 7, mesh ratio - 0.8
```

Do nothing with the remaining settings.

b) The second layer:

```
set size info : height - 0.49, width - 2.5
set material : bulk material - gaas
set doping : doping type - n, concentration - 2e24
set layer mesh : mesh number - 6, mesh ratio - 0.8
```

Do nothing with the remaining settings.

c) The 3rd layer:

```
set size info : height - 0.01, width - 2.5
set material : bulk material - gaas_grading
    • egz - 0
    • composition parameter - 1
    • direction - "+y"
    • grading_from - 0, grading_to - 1
set doping : doping type - n, concentration - 2e24
set layer mesh : mesh number - 5, mesh ratio - 0.8
```

Do nothing with the remaining settings.

d) The 4th layer:

```
set size info : height - 0.05, width - 2.5
set material : bulk material - ingaalp_yt, y - 0.367 , temperature - 300
set doping : doping type - n, concentration - 1e24
set layer mesh : mesh number - 5
    • mesh ratio - 1.5
```

Do nothing with the remaining settings.

e) The 5th layer:

```
set size info : height - 0.95, width - 2.5
set material : bulk material - ingaalp , y - 0.367, temperature - 300
set doping : doping type - n, concentration - 2e24
set layer mesh : mesh number - 9
    • mesh ratio - 0.8
```

Do nothing with the remaining settings.

f) The 6th layer:

set size info : height - 0.1, width - 2.5
set material : bulk material - ingaalp

- composition parameter - 1
- direction - "+y"
- grading_from - 0.364
- grading_to - 0.26

set doping : doping type - none
set layer mesh : mesh number - 6

- mesh ratio - 0.8

Do nothing with the remaining settings.

g) The 7th layer:

set size info : height - 0.01, width - 2.5
set material : bulk material - ingaalp, y - 0.26, temperature - 300
set layer mesh : mesh number - 3

- mesh ratio - 0.8

Do nothing with the remaining settings.

h) The 8th layer:

set size info : height - 0.005, width - 2.5
set material : bulk material - ingaalp_xyt

- x - 0.42
- y - 0
- temperature - 300
- active macro - InGaAlP/GaAs
- xw - 0.42
- yw - 0
- xb - 0.26
- yb - 0.26
- temperature - 300

set layer mesh : mesh number - 3

- mesh ratio - 0.8

Do nothing with the remaining settings.

i) The 9th layer:

set size info : height - 0.005, width - 2.5
set material : bulk material - ingaalp_xyt, x - 0.26, y - 0.26, temperature - 300
set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

j) The 10th layer:

copy the 8th layer and paste here.

k) The 11th layer:

copy the 7th layer and paste here.
set layer mesh : mesh number - 3

- mesh ratio - 1.2

Do nothing with the remaining settings.

l) The 12th layer:

set size info : height - 0.1, width - 2.5

set material : bulk material - ingaalp_yt

- composition parameter - 1
- direction - "+y"
- grading_from - 0.26
- grading_to - 0.364

set layer mesh : mesh number - 6

- mesh ratio - 1.2

Do nothing with the remaining settings.

m) The 13th layer:

set size info : height - 0.65, width - 2.5

set material : bulk material - ingaalp_yt, y - 0.364, temperature - 300

set doping : doping type - p, concentration - 5e23

set layer mesh : mesh number - 9

- mesh ratio - 0.8

Do nothing with the remaining settings.

n) The 14th layer: (click the button "convert to irregular graph")

set size info : height - 0.35, w# - 2.5, upper_w# - 1.8, lower_w# - 2.5

set material : bulk material - ingaalp_yt, y - 0.364, temperature - 300

set doping : doping type - p, concentration - 5e23

set layer mesh : mesh number - 6

- mesh ratio - 1.2

Do nothing with the remaining settings.

o) The 15th layer: (click the button "convert to irregular graph")

set size info : height - 0.05, w# - 2.5, upper_w# - 1.7, lower_w# - 1.8

set material : bulk material - ingaalp_yt, y - 0.364, temperature - 300

set doping : doping type - p, concentration - 2e24

set layer mesh : mesh number - 6

- mesh ratio - 1.2

Do nothing with the remaining settings.

p) The 16th layer:

set size info : height - 0.01, width - 1.7

set material : bulk material - gaas_grading

- egz - 2.4
- composition parameter - 1
- direction - "+y"
- grading_from - 1
- grading_to - 0

set doping : doping type - p, concentration - 1e25

set layer mesh : mesh number - 5, mesh ratio - 1.2, extra_mesh_lower - yes

Do nothing with the remaining settings.

q) The 17th layer:

set size info : height - 0.19, w# - 2.5, upper_w# - 1.5, lower_w# - 1.7

set material : bulk material - gaas

set doping : doping type - p, concentration - 1e25

set contact : side - "top", from - 0, to - 1.5, type - "ohmic"

set layer mesh : mesh number - 6, mesh ratio - 1.2

Do nothing with the remaining settings.

2. Copy the first column to form the second one.

Editing the width of the second column only applies to the current layer, because there are both rectangles and non-rectangular polygons in this column, and the width of the column is not a constant. There are two ways to proceed. One is to edit the width of each layer. The other is to cut the non-rectangular polygon before changing the width of this column, and then add the cut layer. Here the former is used.

a) Modify the bottom layer

set size info : height - 5, width - 7.5

set contact : side - "bottom", from - 0 , to - 7.5, type - "ohmic"

set column mesh : mesh number - 9, mesh ratio - 1.3

Do nothing with the remaining settings.

b) Remove the top contact of the column

c) Change the width from the 2nd layer to 13th layer

set size info : width - 7.5

d) The 14th layer: (click the button "convert to irregular graph")

set size info : height - 0.35, w# - 7.5, upper_w# - 8.2, lower_w# - 7.5

e) The 15th layer:

set size info : height - 0.05, w# - 7.5, upper_w# - 8.3, lower_w# - 8.2

f) The 16th layer:

set size info : height - 0.01, width - 8.3

g) The 17th layer

set size info : height - 0.19, w# - 7.5, upper_w# - 8.5, lower_w# - 8.3

3. Save as "crosslig\lastip_examples\short_wavelength\ingaalp_650nm\ingaalp.layer"

4.5 Example 8

Examples for Apsys: crosslig\apsys_examples\A_tutorial

It is a simple device. Please refer to example

"crosslig\lastip_examples\A_layered_LD." Using short-cut keys and the "drag and release" function will help your design. It consists of 5 layers that use "si" for their "bulk macro" material. P-type doping in the third layer and N-type doping in other layers that form a NPN conjunction.

1. Create the first layer:

set size info : height - 1, width - 1

set material : bulk material - si

set doping : doping type - n, concentration - 1.5e24

set contact : side - "bottom", from - 0 , to - 1, type - "ohmic"

set layer mesh : mesh number - 15, mesh ratio - 0.9

set column mesh : mesh number - 2, mesh ratio - 1

Do nothing with the remaining settings.

2. The 2nd layer:
 - set size info : height - 0.248, width - 1
 - set material : bulk material - si
 - set doping : doping type - n, concentration - 1e2
 - set layer mesh : mesh number - 32, mesh ratio - 0.8
 Do nothing with the remaining settings.
3. The 3rd layer:
 - set size info : height - 0.0039, width - 1
 - set material : bulk material - si
 - set doping : doping type - p, concentration - 2.9e24
 - set layer mesh : mesh number - 15, mesh ratio - 1
 Do nothing with the remaining settings.
4. The 4th layer:
 - set size info : height - 0.0048, width - 1
 - set material : bulk material - si
 - set doping : doping type - n, concentration - 1e2
 - set layer mesh : mesh number - 32, mesh ratio - 1.2
 Do nothing with the remaining settings.
5. The 5th layer:
 - set size info : height - 1, width - 1
 - set material : bulk material - si
 - set doping : doping type - n, concentration - 4e24
 - set contact : side - "top", from - 0, to - 1, type - "ohmic"
 - set layer mesh : mesh number - 15, mesh ratio - 1.1
 Do nothing with the remaining settings.
6. Save as "crosslig\apsys_examples\A_tutorial\camel.layer"

4.6 Example 9

Example for Apsys: crosslig\apsys_examples\Led\gan_led

When you edit the device with large-scale width but small-scale depth, you should modify the picture size by using the short-cut key <alt + s> to fit your view.

1. Create the first column.
 - Look at file "ganled.qw" to get the info for quantum well and barrier.
- a) Create the first layer:
 - set size info : height - 2, width - 400
 - set material : bulk material - gan
 - set doping : doping type - n, concentration - 5e24
 - set layer mesh : mesh number - 5, mesh ratio - 0.8
 - set column mesh : mesh number - 10, mesh ratio - 0.85
 Do nothing with the remaining settings.
- b) The 2nd layer:

set size info : height - 2, width - 400
set material : bulk material - gan
set doping : doping type - n, concentration - $5e24$
set layer mesh : mesh number - 5, mesh ratio - 0.9
Do nothing with the remaining settings.

c) The 3rd layer:
set size info : height - 0.003, width - 400
set material : bulk material - ingan

- x - 0.4
- active macro - InGaN/InGaN
- xw - 0.4
- yw - 0.08
- temperature - 300

set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

d) The 4th layer:
set size info : height - 0.005, width - 400
set material : bulk material - ingan, x - 0.08
set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

e) The 5th layer:
copy the 3rd layer and paste here.

f) The 6th layer:
copy the 4th layer and paste here.

g) The 7th layer:
copy the 3rd layer and paste here.

h) The 8th layer:
copy the 4th layer and paste here.

i) The 9th layer :
copy the 3rd layer and paste here.

j) The 10th layer:
copy the 4th layer and paste here.

k) The 11th layer:
copy the 3rd layer and paste here.

l) The 12th layer:
copy the 4th layer and paste here.

m) The 13th layer:
set size info : height - 0.2, width - 400
set material : bulk material - gan
set doping : doping type - p, concentration - $5e24$
set layer mesh : mesh number - 4, mesh ratio - 1
Do nothing with the remaining settings.

n) The 14th layer:
set size info : height - 0.1, width - 400
set material : bulk material - gan
set doping : doping type - p, concentration - 2e25
set contact : side - "top", from - 0, to - 390, type - "ohmic"
set layer mesh : mesh number - 3, mesh ratio - 1
Do nothing with the remaining settings.

2. Create the second column.
Copy the first column layer and paste here.

a) Modify the first layer:
set size info : height - 2, width - 140
set material : bulk material - gan
set doping : doping type - n, concentration - 1e24
set column mesh : mesh number - 8, mesh ratio - 1.15
Do nothing with the remaining settings.

b) Modify layers from the 2nd layer to the 14th layer.
Because these layers' bulk macro are "void," their active macro should be "void" and their doping should be "none." As a result, these layers are invisible in the drawing area but can be selected and modified.
set material : bulk macro - void, active macro - void
set doping : none

c) Modify the contacts of this column.
select the first layer:
set contact : side - "top", from - 20, to - 140 , type - "ohmic"

3. Save it as file "crosslig\apsys_examples\Led\gan_led\ganled.layer"

An alternative way is to copy the first column to form the second and edit the layers that have different properties.

4.7 Example 10

Examples for Apsys: crosslig\apsys_examples\photo_sensitive\msm155

1. Create the first column of the device

a) Create the first layer:
set size info : height - 2, width - 0.5
set material : bulk material - inp
set doping : doping type - p, concentration - 1e22
set layer mesh : mesh number - 7, mesh ratio - 0.7
set column mesh : mesh number - 5, mesh ratio - 0.7
Do nothing with the remaining settings.

b) The second layer:
set size info : height - 0.05, width - 0.5

set material : bulk material - ingaalas_yt, y - 0.47, temperature - 300
set doping : doping type - p, concentration - 1e22
set layer mesh : mesh number - 5

- mesh ratio - 1

Do nothing with the remaining settings.

c) The 3rd layer :

set size info : height - 0.05, width - 0.5
set material : bulk material - ingaalas_yt

- y - 0
- temperature - 300
- active macro - InGaAlAs
- yw - 0
- temperature - 300

set doping : doping type - p, concentration - 1e22

set layer mesh : mesh number - 9

- mesh ratio - 0.8

Do nothing with the remaining settings.

d) The 4th layer:

set size info : height - 0.05, width - 0.5
set material : bulk material - ingaalas_yt

- y - 0.47
- temperature - 300

set doping : doping type - p, concentration - 1e22

set layer mesh : mesh number - 8

- mesh ratio - 1

contact : "top", from - 0, to - 0.5, contact type - schottky

Do nothing with the remaining settings.

2. Copy the first column to form the second column (column 2). Edit its width, column_mesh, mesh_ratio, top_contact,

a) The first layer:

set size info : height - 2, width - 3.5
set column mesh : mesh number - 21

- mesh ratio - 1.4

Do nothing with the remaining settings.

b) The 4th layer:

set contact : remove top contact

Do nothing with the remaining settings.

3. Copy the first column to form the third column (column 3). Edit its width, column_mesh, mesh_ratio.

a) The first layer:

set column mesh : mesh number - 5, mesh ratio - 0.7

Do nothing with the remaining settings.

4. Save as file "crosslig\apsys_examples\photo_sensitive\msm155\inpmsm.layer"

4.8 Example 11

Example for Pics3d: crosslig\pics3d_examples\A_tutorial

It is a simple device that consists of 6 layers. Follow example crosslig\lastip_examples\A_layered_LD\gaas10 to create the six layers.

1. The first layer:

```
set size info : height - 1, width - 1
set material : bulk material - ingaasp, y - 0
set doping : doping type - n, concentration - 2e24
set layer mesh : mesh number - 10, mesh ratio - 0.8
set column mesh : mesh number - 2, mesh ratio - 1
set contact : side - "bottom", from - 0, to - 1, type - "ohmic"
```

Do nothing with the remaining settings.

2. The second layer:

```
set size info : height - 0.3, width - 1
set material : bulk material - ingaasp, y - 0.22
set doping : doping type - n, concentration - 5e23
set layer mesh : mesh number - 10
    • mesh ratio - "-1.2"
    • shift centre - 0.09504
```

Do nothing with the remaining settings.

3. The third layer

```
set size info : height - 0.2, width - 1
set material : bulk material - ingaasp
    • y - 0.62
    • active macro - InGaAsP
    • yw - 0.62
    • temperature - 300
set layer mesh : mesh number - 11
    • mesh ratio - "-1.2"
```

Do nothing with the remaining settings.

4. The 4th layer:

```
set size info : height - 0.1, width - 1
set material : bulk material - ingaasp, y - 0.22
set doping : doping type - p, concentration - 5e23
set layer mesh : mesh number - 3, mesh ratio - 1.2
```

5. The 5th layer:

```
set size info : height - 0.2, width - 1
set material : bulk material - ingaasp, y - 0.22
set doping : doping type - p, concentration - 5e23
set layer mesh : mesh number - 8, mesh ratio - 0.8
```

Do nothing with the remaining settings.

Comment: Because statement "grating_model" is not available in Layer3d, the parameters of this statement should be added to the .layer file by hand. This function will be implemented in later versions. The parameters of this statement for this device are listed below.

```
grating_model real_index_high= 0.330000E+01 &&  
imag_index_high= 0.000000E+00 &&  
real_index_low= 0.325000E+01 &&  
imag_index_low= 0.000000E+00 &&  
d_high= 0.100000E+00 d_low= 0.100000E+00 &&  
d_fall= 0.000000E+00 d_rise= 0.000000E+00 &&  
grating_xrange=[ 0.000000E+00 0.100000E+01] &&  
grating_yrange=[ 0.160000E+01 0.180000E+01] &&  
grating_order=1
```

6. The 6th layer:

```
set size info : height - 1, width - 1  
set material : bulk material - ingaasp, y - 0  
set doping : doping type - p, concentration - 2e24  
set layer mesh : mesh number - 10, mesh ratio - 1.2  
set contact : side - "top", from - 0, to - 1, type - "ohmic"  
Do nothing with the remaining settings.
```

7. Save it to file "crosslig\pics3d_examples\A_tutorial\inp13.layer"

4.9 Example 12

Examples for Pics3d: crosslig\pics3d_examples\Shuji_3d

1. Create the first column.

a) Create the first layer:

```
set size info : height - 0.3, width - 2  
set material : bulk material - gan  
set doping : doping type - n, concentration - 1e24  
set layer mesh : mesh number - 7, mesh ratio - 1  
set column mesh : mesh number - 4, mesh ratio - 0.85  
set contact : side - "bottom", from - 0, to - 2, type - "ohmic"  
Do nothing with the remaining settings.
```

b) The 2nd layer:

```
set size info : height - 0.1, width - 2  
set material : bulk material - ingan, x- 0.05  
set doping : doping type - n, concentration - 1e24  
set layer mesh : mesh number - 5, mesh ratio - 1
```

c) The 3rd layer:

```
set size info : height - 0.5, width - 2  
set material : bulk material - algan, x- 0.08  
set doping : doping type - n, concentration - 1e24  
set layer mesh : mesh number - 18, mesh ratio - 0.9  
Do nothing with the remaining settings.
```

d) The 4th layer:

set size info : height - 0.1, width - 2
set material : bulk material - gan
set doping : doping type - n, concentration - $1e24$
set layer mesh : mesh number - 5

- mesh ratio - 0.9

Do nothing with the remaining settings.

e) The 5th layer:

set size info : height - 0.0035, width - 2
set material : bulk material - ingan

- x- 0.15
- active macro - InGaN/InGaN
- xw - 0.15
- yw - 0.02
- temperature - 300

set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

f) The 6th layer:

set size info : height - 0.007, width - 2
set material : bulk material - ingan, x- 0.02
set layer mesh : mesh number - 3

- mesh ratio - 1

Do nothing with the remaining settings.

g) The 7th layer:

copy the 5th layer and paste here.

h) The 8th layer:

copy the 6th layer and paste here.

i) The 9th layer:

copy the 5th layer and paste here.

j) The 10th layer:

copy the 6th layer and paste here.

k) The 11th layer:

set size info : height - 0.02, width - 2
set material : bulk material - algan, x- 0.2
set doping : doping type - p, concentration - $3e23$
set layer mesh : mesh number - 6

- mesh ratio - 0.7

Do nothing with the remaining settings.

l) The 12th layer:

set size info : height - 0.1, width - 2
set material : bulk material - gan
set doping : doping type - p, concentration - $5e23$
set layer mesh : mesh number - 9

- mesh ratio - 1.3

Do nothing with the remaining settings.

m) The 13th layer:

set size info : height - 0.5, width - 2
set material : bulk material - algan, x- 0.08
set doping : doping type - p, concentration - 3e23
set layer mesh : mesh number - 18
 • mesh ratio - 1

Do nothing with the remaining settings.

n) The 14th layer:

set size info : height - 0.3, width - 2
set material : bulk material - gan
set doping : doping type - p, concentration - 5e23
set contact : side - "top", from - 0, to - 2, type - "ohmic"
set layer mesh : mesh number - 7
 • mesh ratio - 1

Do nothing with the remaining settings.

2. Copy the first column to form the second column.

a) The first layer:

set column mesh : mesh number - 4, mesh ratio - 1.15
Do nothing with the remaining settings.

b) The top layer:

set material : bulk material - vacuum
set doping : none
set contact : remove "top" contact

Do nothing with the remaining settings.

3. Save it as file "crosslig\pics3d_examples\Shuji_3d\shuji.layer"

